

Linear Filtering - Frequency Domain

$$X_t = \sum_{j=-\infty}^{\infty} b_j Y_{t-j}$$

$$= \sum_{j=-\infty}^{\infty} b_j \left\{ \int_{-\pi}^{\pi} e^{i\omega(t-j)} dS_y(\omega) \right\}$$

$$= \int_{-\pi}^{\pi} e^{i\omega t} \underbrace{\sum_{j=-\infty}^{\infty} b_j e^{-i\omega j}}_{h(\omega)} dS_y(\omega)$$

$$= \int_{-\pi}^{\pi} h(\omega) e^{i\omega t} dS_y(\omega) \quad (*)$$

$$\text{where } h(\omega) = \sum_{j=-\infty}^{\infty} b_j e^{-i\omega j}$$

$$\text{Recall } X_t = \int_{-\pi}^{\pi} e^{i\omega t} dS_x(\omega)$$

$\therefore (*)$ can be written as

$$\int_{-\pi}^{\pi} e^{i\omega t} dS_x(\omega) = \int_{-\pi}^{\pi} h(\omega) e^{i\omega t} dS_y(\omega)$$

$$= \int_{-\pi}^{\pi} e^{i\omega t} [h(\omega) dS_y(\omega)]$$

We deduce that

$$dS_x(\omega) = h(\omega) dS_y(\omega)$$

Lecture 13 : Recall that if $g_x(\omega)$ is the spectral density function of $\{X_t\}$ then

$$\begin{aligned} g_x(\omega) d\omega &= \mathbb{E} [|dS_x(\omega)|^2] \\ &= \mathbb{E} [|h(\omega) dS_y(\omega)|^2] \\ &= |h(\omega)|^2 \mathbb{E} [|dS_y(\omega)|^2] \\ &= |h(\omega)|^2 g_y(\omega) d\omega \end{aligned}$$

$$\therefore g_x(\omega) = |h(\omega)|^2 g_y(\omega)$$